

GLOBAL
EDITION

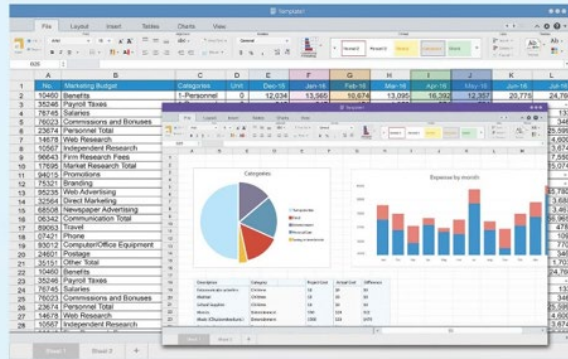


Statistics for Managers

Using Microsoft® Excel®

NINTH EDITION

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Chapter 1

Defining and Collecting Data

Objectives

In this chapter you learn:

- To understand issues that arise when defining variables.
- How to define variables.
- To understand the different measurement scales.
- How to collect data.
- To identify different ways to collect a sample.
- To understand the issues involved in data preparation.
- To understand the types of survey errors.

Why Use Statistics?

- Imagine you are a promotor of a theatrical production in the 1900s. How do you promote and price tickets?
 - Print flyers and place advertisements in local papers.
 - Set a price based on past experience.
 - If the tickets sell out quickly, increase the price next time.
 - If the tickets don't sell, decrease the price next time.
- Jump ahead about 85 years. Start using computer systems to:
 - Sell more categories of tickets, such as premium-priced seats.
 - As customers buy tickets, we monitor sales with schedule reports to add or remove performance dates and customise seat pricing.

Why Use Statistics?

- Jump ahead to today, your fully online ticketing system allows you to:
 - Update seat inventory automatically.
 - Use dynamic pricing to automatically alter seat prices based on factors like peak demand.
 - Gain insights into your customers based on sales data, such as where they live or what other tickets they buy.
 - Knowing your customers allows you to aim advertising and publicity at the correct target market.
 - Using social media, you can determine who is viewing or reacting to your advertising.
- So how effective is this modern approach to advertising?

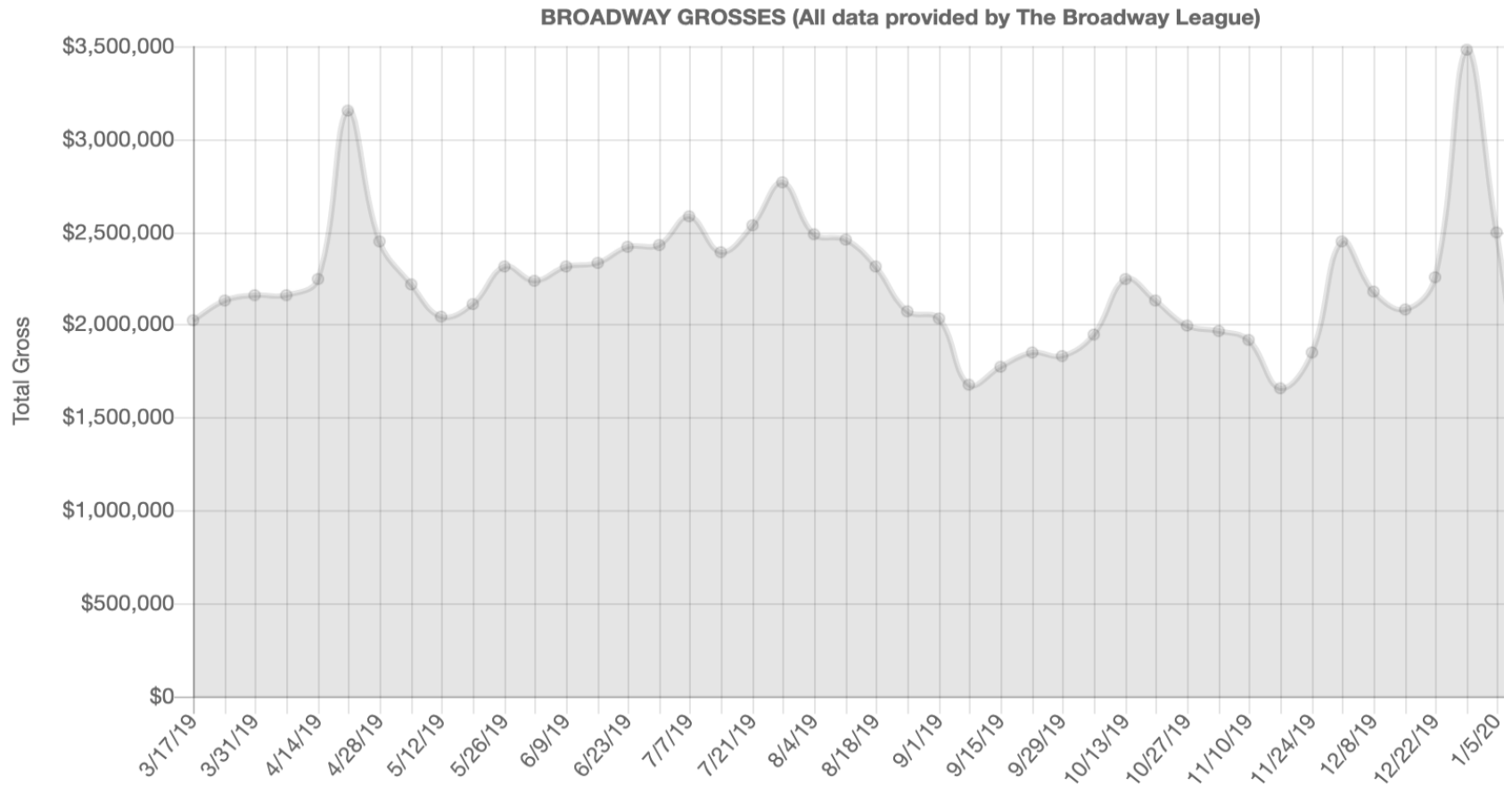
Why Use Statistics?

- The early 2014 financial results showed that *The Lion King* from Disney Theatrical Productions, which aired in 1997, was the top-grossing show on Broadway for 2013.
- This is after the grosses declined by 25% in 2009.
- Four years later, grosses were up 67% and weekly grosses typically exceeded that of the opening weeks of the show, after being adjusted for inflation!
- How did Disney increase sales by 67%? By combining business domain knowledge with *business statistics and analytics* to sell tickets.
- As a musical producer on Broadway said: “We make educated predictions on price, Disney, on the other hand, has turned it into a science.”

Why Use Statistics?

- Disney followed the plan-of-action presented in this course. Disney had:
 - Collected and summarised daily and weekly data.
 - Performed tests and experiments on their data to analyse it.
 - Using the results from these analyses, the insights were used to develop a new interactive seating map that allowed customers to buy tickets for specific seats and permitted Disney to adjust ticket pricing for each seat *and* each performance.
- As a result, *The Lion King* still achieves weekly grosses of around \$2 million after more than 20 years.

Why Use Statistics?



[Link to data \(Paybill\)](#)

Think Differently About Statistics

- Modern-day information technology has allowed businesses to apply statistics in ways that could not be done years ago.
- This course, and prescribed textbook, focusses on business statistics and data science.
- Business statistics emphasises business problem solving and shows a preference to using software to perform calculations.
- All statistical methods require data. **Data** are the facts about the world that one seeks to study and explore.
- Data can be summarised, like weekly or monthly sales.
- Data can be raw (unsummarised), such as the amount or time of a transaction.

What is Statistics?

- **Statistics** is the collection of methods that allow one to work with data effectively.
- Statistics is a TOOL to obtain INFORMATION from DATA.
- It provides us with a formal basis to summarise and visualise data, reach conclusions about the data, make reliable predictions about business activities, and improve the business process.
- Statistics must be applied correctly. Many *professionals* make errors by misusing statistical methods or mistaking statistics as a substitution for, and not an enhancement of, a decision-making process.

A Framework for Statistics

- To minimise errors, we use the DCOVA framework that organises a set of tasks to apply statistics correctly.
- **Define** the data that you want to study to meet an objective.
- **Collect** the data from appropriate sources.
- **Organise** the data collected by developing tables.
- **Visualise** the data by developing charts.
- **Analyse** the data collected, reach conclusions, and present the results.
- Note that the Define and Collect steps must be done before the others. The remaining three are done in varying orders.

Business Analytics

- **Business analytics** combine statistical methods with management science and information systems to form an interdisciplinary tool that supports fact-based decision making. This includes
 - statistical methods to analyse and explore data that can uncover previously unknown or unforeseen relationships.
 - information systems methods to collect and process datasets of all sizes, including very large datasets that would otherwise be hard to use efficiently.
 - management science methods to develop optimisation models that support all levels of management, from strategic planning to daily operations.

Data Science

- **Data Science** is the field of study that combines domain expertise, programming skills, and knowledge of mathematics and statistics to extract meaningful insights from data. Data science practitioners use their methods to:
 - Use a wide range of tools and techniques for evaluating and preparing data.
 - Extract insights from data using predictive analytics and artificial intelligence (AI), including machine learning and deep learning models.
 - Write applications that automate data processing and calculations.
 - Tell and illustrate stories that clearly convey the meaning of results to decision-makers and stakeholders at every level of technical knowledge and understanding.
 - Explain how these results can be used to solve business problems.

Big Data

- In modern statistics, business analytics, or data science, big data plays a vital role.
- **Big data** is a collection of data that cannot be easily browsed or analysed using traditional methods.
- Big data are data being collected in massive volumes, at very fast rates (real time), and in a variety of forms.
- Big data might refer to large datasets of structured data stored in files or worksheets.
- Big data might be **unstructured** such that the data have an irregular pattern and contain values that are not comprehensible without further interpretation.
- **Unstructured data** could be text, pictures, videos, or audio.

Definitions and Terminology

- A **variable** defines a characteristic or property of an item that can vary among the occurrences of those items.
- Using this definition, **data** is a set of values associated with one or more variables.
- Notice that each value for a variable is a single fact – not a list of facts.
- **Statistics** can be defined as the methods that analyse the data of the variables of interest.
- **Descriptive statistics** are the methods of organising, summarising, and presenting data in an informative and convenient way.
- **Inferential statistics** are the methods used to make a conclusion about a characteristic of a population, based on a smaller sample of the population.

Classifying Variables By Type

DCOVA

- **Categorical** (*qualitative*) variables take categories as their values such as “yes”, “no”, or “blue”, “brown”, “green”.
- **Numerical** (*quantitative*) variables have values that represent a counted or measured quantity.
 - **Discrete** variables arise from a *counting process*. Values are **countable** over a finite range.
 - **Continuous** variables arise from a *measuring process*. Values are **uncountable** over a finite range.

Examples of Types of Variables

DCOVA

Question	Responses	Variable Type
Do you have a Facebook profile?	Yes or No	Categorical
How many text messages have you sent in the past three days?	-----	Numerical (discrete)
How long did the mobile app update take to download?	-----	Numerical (continuous)

Measurement Scales

DCOVA

A **nominal scale** classifies categorical data into distinct categories in which no ranking is implied.

<i>Categorical Variables</i>		<i>Categories</i>
Do you have a Facebook profile?	←→	Yes, No
Type of investment	←→	Growth, Value, Other
Cellular Provider	←→	Vodacom, MTN, Cell C, Other, None

Measurement Scales (con't.)

DCOVA

An **ordinal scale** classifies categorical data into distinct categories in which ranking is implied.

<i>Categorical Variable</i>	<i>Ordered Categories</i>
Uber rating for a driver	1 star, 2 stars, ..., 5 stars
Product satisfaction	Very unsatisfied, Fairly unsatisfied, Neutral, Fairly satisfied, Very satisfied
Faculty rank	Associate Professor, Senior Lecturer, Lecturer, Tutor
Standard & Poor's bond ratings	AAA, AA, A, BBB, BB, B, CCC, CC, C, DDD, DD, D
Student Grades	A, B, C, D, F

Measurement Scales (con't.)

DCOVA

Numerical variables use an interval scale or ratio scale.

- An **interval scale** is an ordered scale in which the difference between measurements is a meaningful quantity but the measurements do not have a true zero point.
- A **ratio scale** is an ordered scale in which the difference between the measurements is a meaningful quantity and the measurements have a true zero point.

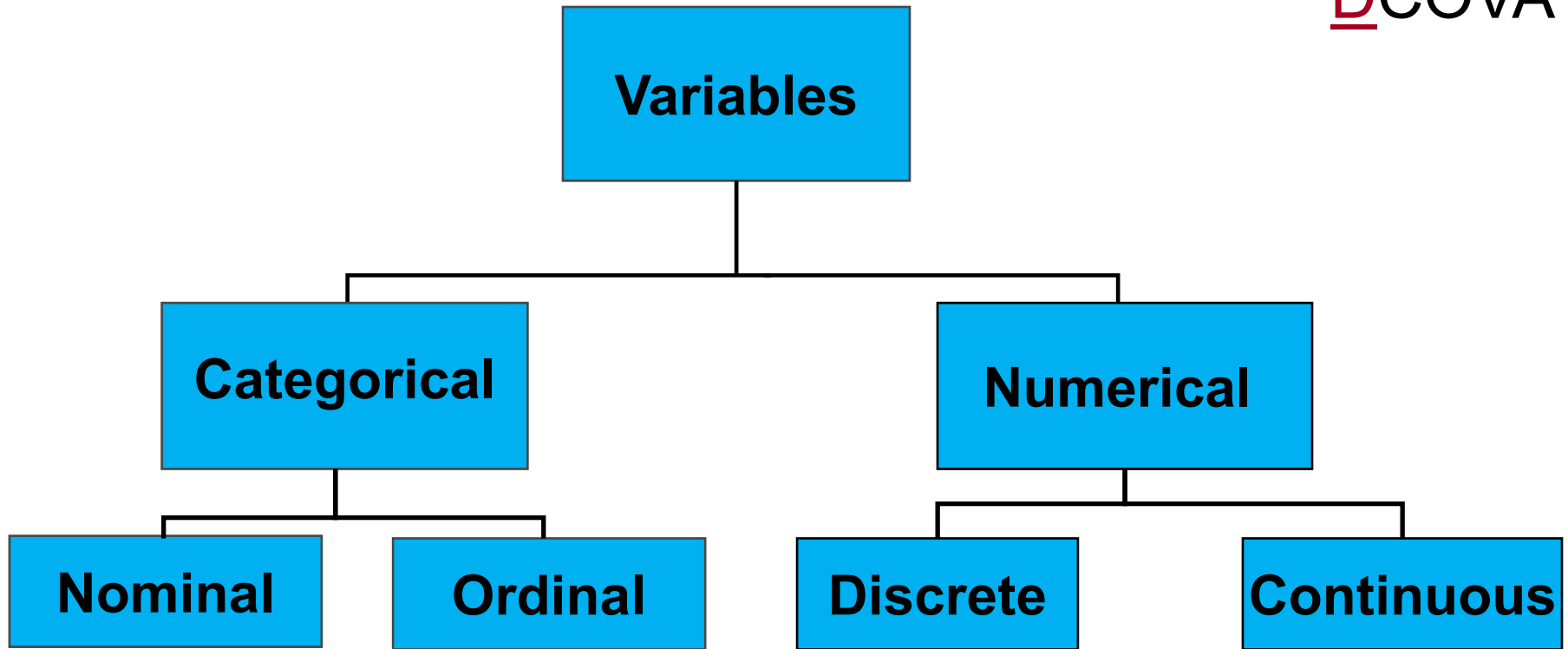
Interval and Ratio Scales

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<i>Numerical Variable</i>		<i>Level of Measurement</i>
Temperature (in degrees Celsius or Fahrenheit)		Interval
Standardized exam score (e.g., ACT or SAT)		Interval
Height (in inches or centimeters)		Ratio
Weight (in pounds or kilograms)		Ratio
Age (in years or days)		Ratio
Salary (in American dollars or Japanese yen)		Ratio

Types of Variables

DCOVA



Examples:

- Marital Status
 - Political Party
 - Eye Color
- (Defined Categories)

Examples: Ratings

- Good, Better, Best
 - Low, Med, High
- (Ordered Categories)

Examples:

- Number of Children
 - Defects per hour
- (Counted items)

Examples:

- Weight
 - Time
- (Measured characteristics)

Data Is Collected From Either A Population or A Sample

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POPULATION

A **population** contains all the items or individuals of interest that you seek to study.

SAMPLE

A **sample** contains only a portion of a population of interest.

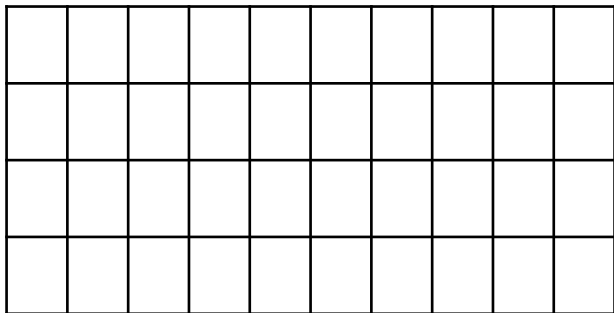
Population vs. Sample

DCOVA

Population

All the items or individuals about which you want to reach conclusion(s).

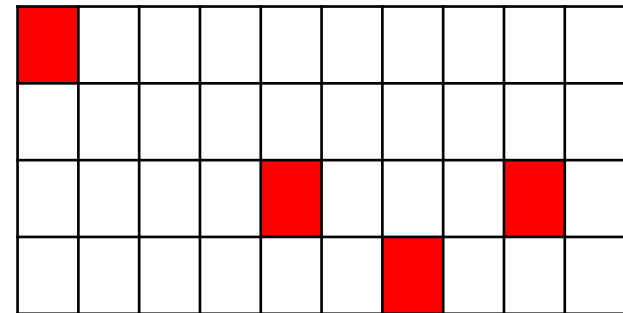
A Population of Size 40



Sample

A portion of the population of items or individuals.

A Sample of Size 4



Why Use Sampling?

DCOVA

- Less time consuming than selecting every item in the population.
- Less costly than selecting every item in the population.
- Less cumbersome and more practical than analysing the entire population.

A Sample Is Analysed To Estimate Characteristics Of An Entire Population

DCOVA

- A **population parameter** summarises the value of a specific variable for a population.
- A **sample statistic** summarises the value of a specific variable for sample data.
- **Sample statistics** are used to estimate **population parameters**.

Sources Of Data Arise From The Following Activities

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- Capturing data generated by ongoing business activities.
- Distributing data compiled by an organisation or individual.
- Compiling the responses from a survey.
- Conducting a designed experiment and recording the outcomes.
- Conducting an observational study and recording the results.

Examples of Data Collected From Ongoing Business Activities

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- A bank studies years of financial transactions to help them identify patterns of fraud.
- Economists use data on searches done via Google to help forecast future economic conditions.
- Marketing companies use tracking data to evaluate the effectiveness of a web site.

Examples Of Data Distributed By An Organisation or Individual

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- Financial data on a company provided by investment services.
- Industry or market data from market research firms and trade associations.
- Stock prices, weather conditions, and sports statistics in daily newspapers.

Examples of Survey Data

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- A survey asking people which laundry detergent has the best stain-removing abilities.
- Political polls of registered voters during political campaigns.
- People being surveyed to determine their satisfaction with a recent product or service experience.

Examples of Data From A Designed Experiment

DCOVA

- Consumer testing of different versions of a product to help determine which product should be pursued further.
- Material testing to determine which supplier's material should be used in a product.
- Market testing on alternative product promotions to determine which promotion to use more broadly.

Examples of Data Collected From Observational Studies

DCOVA

- Market researchers using focus groups to elicit unstructured responses to open-ended questions.
- Measuring the time it takes for customers to be served in a fast food establishment.
- Measuring the volume of traffic through an intersection to determine if some form of advertising at the intersection is justified.

Observational Studies & Designed Experiments Have A Common Objective

DCOVA

- Both are attempting to quantify the effect that a process change (called a **treatment**) has on a variable of interest.
- In an observational study, there is no direct control over which items receive the treatment.
- In a designed experiment, there is direct control over which items receive the treatment.

Sources of Data

DCOVA

- **Primary Sources:** The data collector is the one using the data for analysis:
 - Data from a political survey.
 - Data collected from an experiment.
 - Observed data.
- **Secondary Sources:** The person performing data analysis is not the data collector:
 - Analysing census data.
 - Examining data from print journals or data published on the Internet.

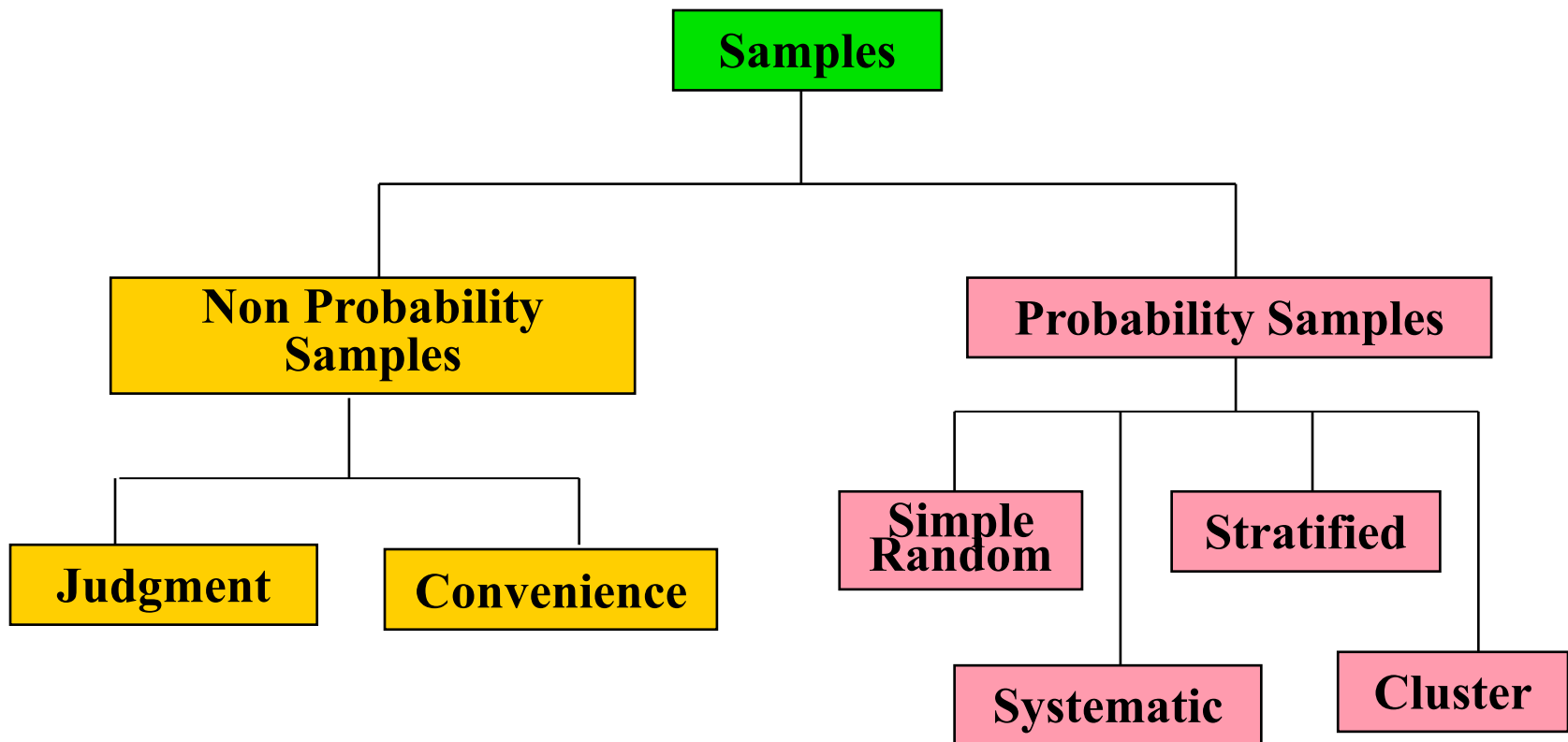
A Sampling Process Begins With A Sampling Frame

DCOVA

- The sampling frame is a listing of items that make up the population.
- Frames are data sources such as population lists, directories, or maps.
- Inaccurate or biased results can result if a frame excludes certain groups or portions of the population.
- Using different frames to generate data can lead to dissimilar conclusions.

Types of Samples

DCOVA



Types of Samples:

Nonprobability Sample

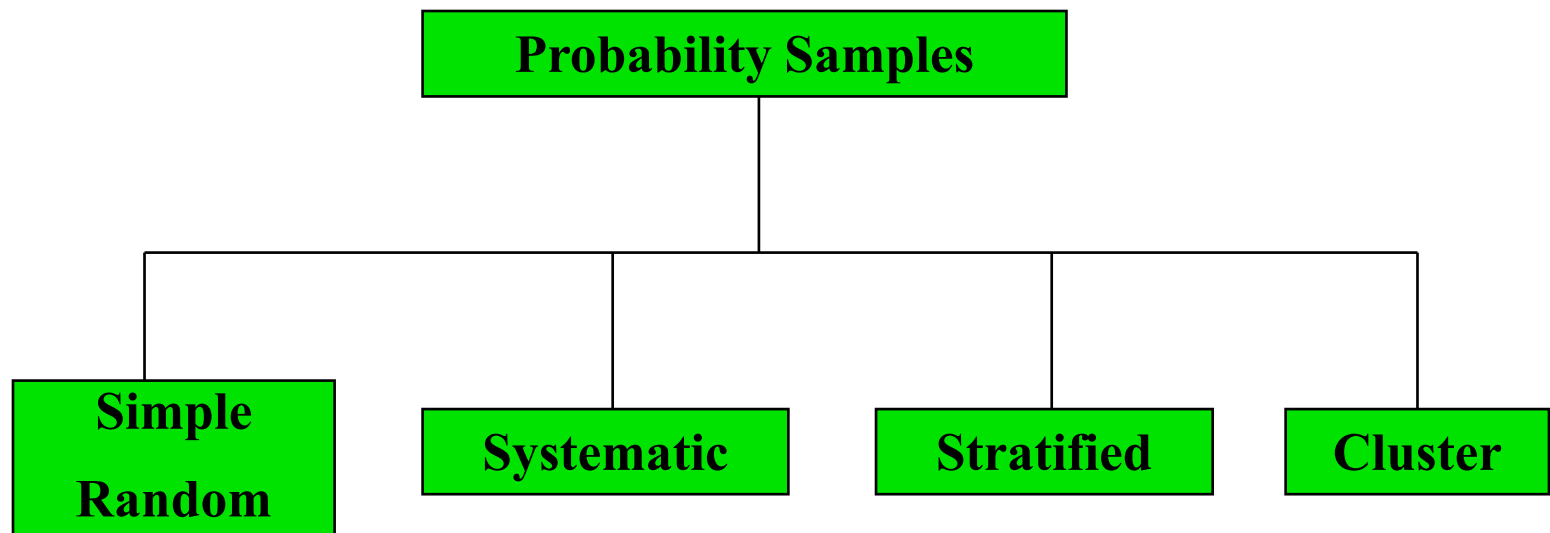
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- In a nonprobability sample, items included are chosen without regard to their probability of occurrence.
 - In **convenience sampling**, items are selected based only on the fact that they are easy, inexpensive, or convenient to sample.
 - In a **judgment sample**, you get the opinions of pre-selected experts on the subject matter.

Types of Samples: Probability Sample

DCOVA

- In a **probability sample**, items in the sample are chosen on the basis of known probabilities.



Probability Sample: Simple Random Sample

DCOVA

- Every individual or item from the frame has an equal chance of being selected.
- Selection may be with replacement (selected individual is returned to frame for possible reselection) or without replacement (selected individual is not returned to the frame).
- Samples obtained from table of random numbers or computer random number generators.

Selecting a Simple Random Sample Using A Random Number Table

DCOVA

Sampling Frame For Population With 850 Items

<u>Item Name</u>	<u>Item #</u>
Bev R.	001
Ulan X.	002
.	.
.	.
.	.
.	.
Joann P.	849
Paul F.	850

Portion Of A Random Number Table

49280 88924 35779 00283 81163 07275
11100 02340 12860 74697 96644 89439
09893 23997 20048 49420 88872 08401

Select item as the number corresponding to three digits in the random number table.

The First 5 Items in a simple random sample

Item # 492

Item # 808

Item # 892 -- does not exist so ignore

Item # 435

Item # 779

Item # 002

Probability Sample: Systematic Sample

DCOVA

- Decide on sample size: n
- Divide frame of N individuals into groups of k individuals: $k = N/n$
- Randomly select one individual from the 1st group, i.e. choose a sample between 1 and k
- Select every k^{th} individual thereafter

$$N = 40$$

$$n = 4$$

$$k = 10$$

First Group

First Value

Probability Sample: Stratified Sample

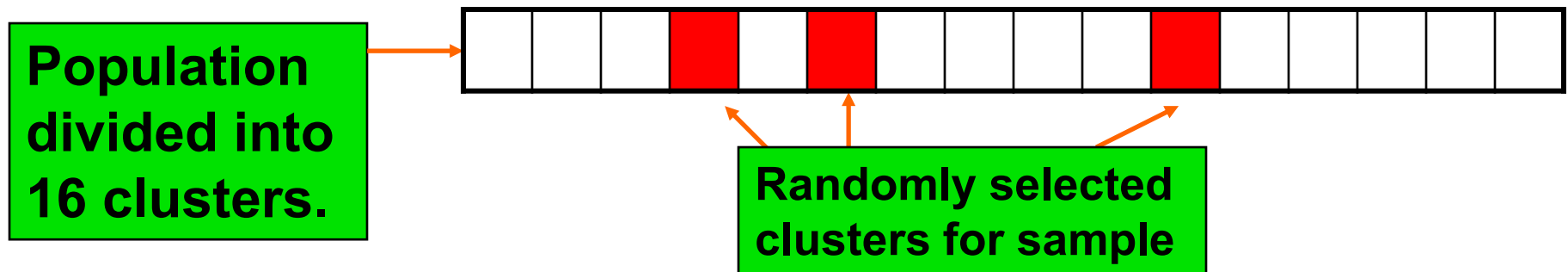
DCOVA

- Divide population into two or more subgroups (called *strata*) according to some common characteristic.
- A simple random sample is selected from each subgroup, with sample sizes proportional to strata sizes.
- Samples from subgroups are combined into one.
- This is a common technique when sampling population of voters, stratifying across provincial or socio-economic lines.

Probability Sample Cluster Sample

DCOVA

- Population is divided into several “clusters,” each representative of the population.
- A simple random sample of clusters is selected.
- All items in the selected clusters can be used, or items can be chosen from a cluster using another probability sampling technique.
- A common application of cluster sampling involves election exit polls, where certain election districts are selected and sampled.



Probability Sample: Comparing Sampling Methods

DCOVA

- Simple random sample and Systematic sample:
 - Simple to use.
 - May not be a good representation of the population's underlying characteristics.
- Stratified sample:
 - Ensures representation of individuals across the entire population.
- Cluster sample:
 - More cost effective.
 - Less efficient (need larger sample to acquire the same level of precision).

Probability Sample: Selection with Probability Proportionate To Size

DCOVA

- In the case of a random sample, elements of the population (e.g. invoices) are selected without the monetary value on the invoice playing a role.
- For example, if we consider sales, an invoice with an entry of R10 has the same probability of being selected as an invoice with an entry of R1 000.
- If the correctness of a monetary value must be verified, the magnitude of the monetary value becomes important.

Selection with PPS

DCOVA

- In such a case a selection process that takes the magnitude of monetary values on each invoice into account, is preferred.
- We refer to this type of selection process as selection proportional to size, where size refers to the monetary value on each invoice.
- Suppose several invoices must be selected from N invoices via the probability proportional to size (PPS) selection process.

Selection with PPS

DCOVA

- Let T denote the total monetary value (measured in rand) of the N invoices.
- According to the PPS selection process, each of the T rand units has an equal probability of being selected.
- This implies that an invoice with a R4 000 entry has a probability of selection four times as large as the selection probability of an invoice with a R1 000 entry.

Selection with PPS

DCOVA

- From there the phrase *PROBABILITY PROPORTIONAL TO SIZE*.
- It is important to note that in this case we are dealing with two types of elements, invoices and rand units.
- With PPS selection an invoice is selected in an indirect manner, because a rand unit is selected first and then the invoice on which it occurs is selected.

Selection with PPS

DCOVA

Note that each rand unit has the same chance of selection, but the chance of selection for each invoice is proportionate to the number of rand units that appears on it.

Selection with PPS: Example

DCOVA

Suppose the total monetary value of sales on $N = 1500$ invoices is R225 000 and that $n = 20$ invoices must be selected according to a PPS selection process. First, generate twenty 6-digit random numbers between 000001 and 225000 and arrange these numbers in ascending order.

Then use the following table to determine which invoices (which could be less than 20 if some invoices were selected more than once) to associate with the random numbers:

Invoice	Monetary Value (R)	Accumulated value (R)	Interval Values	6-Digit Random Numbers
1	10			
2	112			
3	78			
4	150			
5	5872			
6	613			
7	114			
8	14			
etc	etc	etc	etc	etc

Selection with PPS

DCOVA

Suppose the first four 6-digit random numbers are 000098, 000517, 000972 and 006863.

The number 000098 lies between 11 and 122, therefore invoice 2 is selected; 000517 lies between 351 and 6222, therefore invoice 5 is selected.

Selection with PPS

DCOVA

The number 000972 also indicates invoice 5, but note that no invoice may be selected twice.

Moreover, 006863 lies between 6 836 and 6 949, therefore invoice 7 is selected.

Note that the selection probability of invoice 1 is $10/225\ 000$, while the selection probability of invoice 5 is $5872/225000$, etc.

Invoice	Monetary Value (R)	Accumulated value (R)	Interval Values	6-Digit Random Numbers
1	10	10	01 - 10	
2	112	122	11-122	000098
3	78	200	123-200	
4	150	350	201-350	
5	5872	6222	351-6222	000517 & 000972
6	613	6835	6223-6835	
7	114	6949	6836-6949	006863
8	14	6963	6950-6963	
etc	etc	etc	etc	etc

Data Cleaning Is An Important Data Preprocessing Task Prior To Analysis

DCOVA

Data cleaning corrects irregularities in the data:

- Invalid variable values, including:
 - Non-numerical data for numerical variable.
 - Invalid categorical values for a categorical variable.
 - Numeric values outside a defined range.
- Coding errors, including:
 - Inconsistent categorical values.
 - Inconsistent case for categorical values.
 - Unrelated / Unwanted characters.
- Data integration errors, including:
 - Redundant columns.
 - Duplicated rows.
 - Differing column lengths.
 - Different units of measure or scale for numerical variables.

Cleaning Invalid Variable Values Can Be Semi-Automated

DCOVA

- Invalid variable values can be identified by simple scanning techniques, for example:
 - Non-numeric entries for numerical variables.
 - Values for categorical variables that don't match a pre-defined category.
 - Values for a numeric variable outside a pre-defined explicit range.
- Features exist in Excel to assist in this task.

Examples Of Coding Errors

DCOVA

Copy-and-paste or data import can result in poor recording or entry of data.

Categorical variable: Gender, Correct coding: F or M

- Correctable error: Female.
- Invalid data: New York.
- Correctable or software tolerated: m.
- Irregular and nonprintable (hidden) characters:
 - Leading or trailing space(s): _F or F_.
 - Other nonprintable characters may also be leading or trailing

Data Integration Errors From Combining Two Different Computerised Data Sources

DCOVA

- Data integration errors often requires time-consuming manual effort.
- Some examples:
 - Variable names or definitions may differ.
 - Duplicated rows (observations) may also occur.
 - Different units of measurement (or scale) may not be obvious without human interpretation.

Data Can Be Formatted and / or Encoded In More Than One Way

DCOVA

- Some electronic formats are more readily usable than others.
- Different encodings can impact the precision of numerical variables and can also impact data compatibility.
- As you identify and choose sources of data you need to consider / deal with these issues.

Missing Values Are Values Not Collected For A Variable

DCOVA

- Survey data may include answers for which no response was given by the survey taker.
- Missing values can also result from integrating two data sources.
- Do not confuse missing values with miscoded values.

Data Cleaning Cannot Be A Fully Automated Process

DCOVA

- Excel and Tableau have functionality to lessen the burden of data cleaning.
- The software guides in the course textbook explain this functionality.
- When performing data cleaning, always preserve a copy of the original data for later reference.

Other Data Preprocessing Tasks

DCOVA

■ Data Formatting

- Rearranging data structure or changing electronic encoding of the data or both.

■ Stacking and Unstacking Data

- Analysis of a numerical variable may require subdividing that data into two or more groups.
- Unstacking involves creating separate numerical variables for the different groups.
- Stacking involves pairing the one numerical variable with a second categorical variable.

■ Recoding Variables

- Redefining categories for a categorical variable.
- Transforming a numerical variable into a categorical variable.

Recoding Of Variables

DCOVA

- Recoding a variable can either supplement or replace the original variable.
- Recoding a categorical variable involves redefining categories.
- Recoding a numerical variable involves changing this variable into a categorical variable.
- When recoding be sure that the new categories are mutually exclusive (categories do not overlap) and collectively exhaustive (categories cover all possible values).

Evaluating Survey Worthiness

DCOVA

- What is the purpose of the survey?
- Is the survey based on a probability sample?
- Coverage error – appropriate frame?
- Nonresponse error – follow up.
- Measurement error – good questions elicit good responses.
- Sampling error – always exists.

Types of Survey Errors

DCOVA

- Coverage error or selection bias:
 - Exists if some groups are excluded from the frame and have no chance of being selected.
- Nonresponse error or bias:
 - People who do not respond may be different from those who do respond.
- Sampling error:
 - Variation from sample to sample **will always exist**.
- Measurement error:
 - Due to weaknesses in question design and / or respondent error.

Types of Survey Errors *(continued)*

DCOVA

- Coverage error

**Excluded from
frame**

- Nonresponse error

**Follow up on
nonresponses**

- Sampling error

**Random
differences from
sample to sample**

- Measurement error

**Bad or leading
question**

Ethical Issues About Surveys

DCOVA

- Coverage error and nonresponse error can be leveraged by survey designers to purposely bias survey results.
- Sampling error can be an ethical issue if the findings are purposely not reported with the associated margin of error.
- Measurement error can be an ethical issue:
 - Survey sponsor chooses leading questions.
 - Interviewer purposely leads respondents in a particular direction.
 - Respondent(s) wilfully provide false information.

Excel: Sampling

- We use the Data Analysis tool in Excel to obtain a simple random sample with replacement or a systematic sample.
- We do not cover Excel implementations of sampling without replacement, stratified sampling or cluster sampling.
- The population data are structured in a table where each column is a variable, and each row is an observation of these variables.
- Notice we need a column that specifies the observation number (index) in the population data frame.

Excel: Sampling

	A	B	C
1	Population Data Frame		
2	Row Index	X1	X2
3	1	20	A
4	2	149	B
5	3	355.5	A
6	4	219.5	B
7	5	308.5	B
8	6	340	A

We collect data on two variables, $X1$ and $X2$.

The population consists of six observations.

$X1$ is a numerical variable and $X2$ is a categorical variable.

Excel: Sampling

The following steps are used to perform a simple random sample with replacement of the row index.

1. Make sure the Data Analysis tool is installed.
2. Click the **Data** ribbon and select **Data Analysis**.
3. Select **Sampling** and click **OK**.
4. Set the **Input Range** equal to A3:A8.
5. Since the input range does not contain a column name, make sure **Labels** is unchecked.
6. Set the **Sampling Method** to **Random**, and type **4** in the **Number of Samples** box.
7. Under **Output options**, click the **Output Range** option and type **E3**.
8. Click **OK**.

Excel: Sampling

- After running the Sampling tool, Excel returns four random values selected from the **Row Index** column in cells E3 to E6.
- We must now return the values for *X1* and *X2* corresponding to these row indexes.
- The output should look as follows:

	A	B	C	D	E	F	G
1	Population Data Frame						
2	Row Index	X1	X2			X1	X2
3	1	20	A		6		
4	2	149	B		3		
5	3	355.5	A		2		
6	4	219.5	B		3		
7	5	308.5	B				
8	6	340	A				
9							

Excel: Sampling

- We notice that the Sampling tool selected the row indices 6, 3, 2, and 3 from the population data frame.
- To return the corresponding values for $X1$ and $X2$, we use the INDEX() function.
- For example, the first observation sampled from the population is row 6. Here $X1 = 340$ and $X2 = A$.
- To return the 340 in cell F3, type the following in cell F3: =INDEX(\$B\$3:\$B\$8, E3).
- This function first looks at the value of E3 which is 6. It then searches and returns the 6th value in the range B3:B8. The dollar signs “\$” ensure that this range remains fixed if we move the formula.

Excel: Sampling

- This is done in each following row in column F by dragging this function down to cell F6. Notice that the second entry for the INDEX function changes with each row.

E	F	G
	X1	X2
6	=INDEX(\$B\$3:\$B\$8,E3)	
3	=INDEX(\$B\$3:\$B\$8,E4)	
2	=INDEX(\$B\$3:\$B\$8,E5)	
3	=INDEX(\$B\$3:\$B\$8,E6)	

E	F	G
	X1	X2
6	340	
3	355.5	
2	149	
3	355.5	

- Next, we apply the same function to column G. Notice that we change the range to column C.

Excel: Sampling

Sampled data:

E	F	G
	X1	X2
6	340	A
3	355.5	A
2	149	B
3	355.5	A

Excel formulas:

E	F	G
	X1	X2
6	=INDEX(\$B\$3:\$B\$8,E3)	=INDEX(\$C\$3:\$C\$8,E3)
3	=INDEX(\$B\$3:\$B\$8,E4)	=INDEX(\$C\$3:\$C\$8,E4)
2	=INDEX(\$B\$3:\$B\$8,E5)	=INDEX(\$C\$3:\$C\$8,E5)
3	=INDEX(\$B\$3:\$B\$8,E6)	=INDEX(\$C\$3:\$C\$8,E6)

Excel: Sampling

- The Sampling tool in the Data Analysis tool can be used for a systematic sample.
- Go Data -> Data Analysis -> Sampling and click OK.
- Choose $k = 2$ for a systematic sample by selecting **Periodic** under **Sampling Method** and set the **Period** equal to 2.
- In the **Output Range** dialog click cell A10.
- This produces a systematic sample of the row indices of the population data frame.
- As before, we can return the corresponding values for $X1$ and $X2$ using INDEX().

Excel: Sampling

Note that we can use INDEX(array, row num) or VLOOKUP(lookup value, table array, column index) to return X1 and X2.

	A	B	C	D	E	F	G
1	Index	X1	X2		Return Index	X1 using INDEX()	X2 using VLOOKUP()
2	1	-5	A		4	19	D
3	2	24	B		1	-5	A
4	3	3	C				
5	4	19	D				
6	5	10	E				

	A	B	C	D	E	F	G
1	Index	X1	X2		Return Index	X1 using INDEX()	X2 using VLOOKUP()
2	1	-5	A		4	=INDEX(\$B\$2:\$B\$6, E2)	=VLOOKUP(E2,\$A\$2:\$C\$6,3)
3	2	24	B		1	=INDEX(\$B\$2:\$B\$6, E3)	=VLOOKUP(E3,\$A\$2:\$C\$6,3)
4	3	3	C				
5	4	19	D				
6	5	10	E				

Chapter Summary

In this chapter we have discussed:

- Understanding issues that arise when defining variables.
- How to define variables.
- Understanding the different measurement scales.
- How to collect data.
- Identifying different ways to collect a sample.
- Understanding the issues involved in data preparation.
- Understanding the types of survey errors.